

UNIVERSITY OF THE ARMED FORCES - ESPE

Department of Exact Sciences

Object - Oriented Programming

NRC: 368663

WS15 - CollectionsExamples

LASCANO, JORGE EDISON

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Workshop 15 – Collections Examples

Topic: Collections Implementation in Java, Python, and TypeScript

1. Java Implementation

1.1. Part 1 – Collection Initialization

1.1.1. Source Code

```java
package ec.edu.espe.collections;

import java.util.ArrayList;
import java.util.Collection;
import ec.edu.espe.model.Person;

/**
 *
 * @author Dylan Cumbajin, MosQ, @ESPE
 */
public class MyThings {
    public static void main(String[] args) {
        System.out.println("====================");
        System.out.println("| Collections ws   |");
        System.out.println("| Dylan Cumbajin  |");
        System.out.println("====================");

        Person person;
        Collection things;
        things = new ArrayList();
        person = new Person(1, "Isabel", 300.25F);

        System.out.println("sizze of my collection at the beginning -->"+things.size());

        things.add(18000);
        things.add("Heloo OOP Programmers from Dylan");
        things.add(4073.78F);
        things.add(person);

        System.out.println("sizze of my collection now -->"+things.size());
        System.out.println("my ArrayList elemnts are --> " + things);

```

1.1.2. Execution Results

```
run:
====================
| Collections ws   |
| Dylan Cumbajin  |
====================
sizze of my collection at the beginning -->0
sizze of my collection now -->4
my ArrayList elemnts are --> [18000, Heloo OOP Programmers from Dylan, 4073.78, id=1, name=Isabel, salary=300.25]
sizze of my collection now -->3
my ArrayList elemnts are --> [18000, Heloo OOP Programmers from Dylan, id=1, name=Isabel, salary=300.25]
====================
```

1.2. Part 2 – Collection Operations

1.2.1. Source Code

```
36
37          things.remove(4073.78F);
38          System.out.println("sizze of my collection now -->"+things.size());
39          System.out.println("my ArrayList elemnts are --> " + things);
40
41          System.out.println("==================");
42
43          Collection<Person> persons;
44          persons = new ArrayList<>();
45          persons.add(person);
46          person = new Person(2, "Martin Onate", 4000.35F);
47          persons.add(person);
48          person = new Person(3, "Erick Revelo", 480.87F);
49
50          persons.add(person);
            person = new Person(4, "Nestor Olivo", 4480.89F);
52
53                  System.out.println("persons --> " + persons);
54          System.out.println("==================");
55
56          persons.forEach((p) -> {
57              System.out.println("Person --> " + p);
58          });
59
60          System.out.println("==================");
61          for (Person p : persons) {
62              System.out.println("P --> " + p);
63          }
64
65          for (int i = 0; i < 10; i++) {
66              persons.add(new Person(i + 1, "Edison" + (i + 1), (i + 1) * 1000));
67          }
68
69          System.out.println("size of persons ArrayList --> " + persons.size());
70          persons.forEach((p) -> {
71              System.out.println("Person --> " + p);
72          });
73
74      }
75
76  }
77
```

1.2.2. Execution Results

```
==================
persons --> [id=1, name=Isabel, salary=300.25, id=2, name=Martin Onate, salary=4000.35, id=3, name=Erick Revelo, salary=480.87]
==================
Person --> id=1, name=Isabel, salary=300.25
Person --> id=2, name=Martin Onate, salary=4000.35
Person --> id=3, name=Erick Revelo, salary=480.87
==================
P --> id=1, name=Isabel, salary=300.25
P --> id=2, name=Martin Onate, salary=4000.35
P --> id=3, name=Erick Revelo, salary=480.87
size of persons ArrayList --> 13
Person --> id=1, name=Isabel, salary=300.25
Person --> id=2, name=Martin Onate, salary=4000.35
Person --> id=3, name=Erick Revelo, salary=480.87
Person --> id=1, name=Edison1, salary=1000.0
Person --> id=2, name=Edison2, salary=2000.0
Person --> id=3, name=Edison3, salary=3000.0
Person --> id=4, name=Edison4, salary=4000.0
Person --> id=5, name=Edison5, salary=5000.0
Person --> id=6, name=Edison6, salary=6000.0
Person --> id=7, name=Edison7, salary=7000.0
Person --> id=8, name=Edison8, salary=8000.0
Person --> id=9, name=Edison9, salary=9000.0
Person --> id=10, name=Edison10, salary=10000.0
BUILD SUCCESSFUL (total time: 0 seconds)
```

2. Python Implementation

2.1. Part 1 – Collection Initialization

2.1.1. Source Code

```python
"""
@author Dylan Cumbajin, MosQ, @ESPE
"""

def main():
    print("====================")
    print("| Collections ws   |")
    print("| Dylan Cumbajin   |")
    print("====================")

    things = []
    person = Person(1, "Isabel", 300.25)

    print("sizze of my collection at the beginning -->", len(things))
```

2.1.2. Execution Results

```
====================
| Collections ws   |
| Dylan Cumbajin   |
====================
sizze of my collection at the beginning --> 0
sizze of my collection now --> 4
my ArrayList elemnts are --> [18000, 'Heloo OOP Programmers from Dylan', 4073.78, id=1, name=Isabel, salary=300.25]
sizze of my collection now --> 3
my ArrayList elemnts are --> [18000, 'Heloo OOP Programmers from Dylan', id=1, name=Isabel, salary=300.25]
====================
```

2.2. Part 2 – Collection Operations

2.2.1. Source Code

```
  things.append(18000)
    things.append("Heloo OOP Programmers from Dylan")
    things.append(4073.78)
    things.append(person)

    print("sizze of my collection now -->", len(things))
    print("my ArrayList elemnts are -->", things)

    things.remove(4073.78)

    print("sizze of my collection now -->", len(things))
    print("my ArrayList elemnts are -->", things)

    print("====================")

    persons = []
    persons.append(person)

    person = Person(2, "Martin Onate", 4000.35)
    persons.append(person)

    person = Person(3, "Erick Revelo", 480.87)
    persons.append(person)

    person = Person(4, "Nestor Olivo", 4480.89)

    print("persons -->", persons)
    print("==================")

    for p in persons:
        print("Person -->", p)

    print("==================")

    for p in persons:
        print("P -->", p)

    for i in range(10):
        persons.append(
            Person(i + 1, "Edison" + str(i + 1), (i + 1) * 1000)
        )

    print("size of persons ArrayList -->", len(persons))

    for p in persons:
        print("Person -->", p)


if __name__ == "__main__":
    main()
```

2.2.2. Execution Results

```
my ArrayList elemnts are --> [18000, 'Heloo OOP Programmers from Dylan', id=1, name=Isabel, salary=300.25]
==================
persons --> [id=1, name=Isabel, salary=300.25, id=2, name=Martin Onate, salary=4000.35, id=3, name=Erick Revelo, salary=480.87]
==================
Person --> id=1, name=Isabel, salary=300.25
Person --> id=2, name=Martin Onate, salary=4000.35
Person --> id=3, name=Erick Revelo, salary=480.87
=================
P --> id=1, name=Isabel, salary=300.25
P --> id=2, name=Martin Onate, salary=4000.35
P --> id=3, name=Erick Revelo, salary=480.87
size of persons ArrayList --> 13
Person --> id=1, name=Isabel, salary=300.25
Person --> id=2, name=Martin Onate, salary=4000.35
Person --> id=3, name=Erick Revelo, salary=480.87
Person --> id=1, name=Edison1, salary=1000
Person --> id=2, name=Edison2, salary=2000
Person --> id=3, name=Edison3, salary=3000
Person --> id=4, name=Edison4, salary=4000
Person --> id=5, name=Edison5, salary=5000
Person --> id=6, name=Edison6, salary=6000
Person --> id=7, name=Edison7, salary=7000
Person --> id=8, name=Edison8, salary=8000
Person --> id=9, name=Edison9, salary=9000
Person --> id=10, name=Edison10, salary=10000
dy456 ~\Desktop\ESPE2609-OOPSW36863\ws\cumbajin\U1\ws15CollectionsExamples\ws15CollectionsPython main ≡*?2 -8 | ~1
❯
```

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3. TypeScript Implementation

3.1. Part 1 – Collection Initialization

3.1.1. Source Code

```typescript
/**
 * @author Dylan Cumbajin, MosQ, @ESPE
 */
class MyThings {

    static main(): void {
        console.log("====================");
        console.log("| Collections ws   |");
        console.log("| Dylan Cumbajin  |");
        console.log("====================");

        let person: Person;
        let things: any[];

        things = [];
        person = new Person(1, "Isabel", 300.25);
```

3.1.2. Execution Results

```
====================
| Collections ws   |
| Dylan Cumbajin   |
====================
sizze of my collection at the beginning -->0
sizze of my collection now -->4
my ArrayList elemnts are --> [
  18000,
  'Heloo OOP Programmers from Dylan',
  4073.78,
  Person { id: 1, name: 'Isabel', salary: 300.25 }
]
sizze of my collection now -->3
my ArrayList elemnts are --> [
  18000,
  'Heloo OOP Programmers from Dylan',
  Person { id: 1, name: 'Isabel', salary: 300.25 }
]
```

3.2. Part 2 – Collection Operations

3.2.1. Source Code

```
things.push("Heloo OOP Programmers from Dylan");
        things.push(4073.78);
        things.push(person);

        console.log("sizze of my collection now -->" + things.length);
        console.log("my ArrayList elemnts are -->", things);

        things.splice(things.indexOf(4073.78), 1);

        console.log("sizze of my collection now -->" + things.length);
        console.log("my ArrayList elemnts are -->", things);

        console.log("==================");

        let persons: Person[];
        persons = [];

        persons.push(person);

        person = new Person(2, "Martin Onate", 4000.35);
        persons.push(person);

        person = new Person(3, "Erick Revelo", 480.87);
        persons.push(person);

        person = new Person(4, "Nestor Olivo", 4480.89);

        console.log("persons -->", persons);
        console.log("==================");

        persons.forEach((p) => {
            console.log("Person --> " + p.toString());
        });

        console.log("==================");

        for (let p of persons) {
            console.log("P --> " + p.toString());
        }

        for (let i = 0; i < 10; i++) {
            persons.push(
                new Person(i + 1, "Edison" + (i + 1), (i + 1) * 1000)
            );
        }

        console.log("size of persons ArrayList -->" + persons.length);

        persons.forEach((p) => {
            console.log("Person --> " + p.toString());
        });
    }
}

MyThings.main();
```

3.2.2. Execution Results

```
sizze of my collection now -->3
my ArrayList elemnts are --> [
  18000,
  'Heloo OOP Programmers from Dylan',
  Person { id: 1, name: 'Isabel', salary: 300.25 }
]
====================
persons --> [
  Person { id: 1, name: 'Isabel', salary: 300.25 },
  Person { id: 2, name: 'Martin Onate', salary: 4000.35 },
  Person { id: 3, name: 'Erick Revelo', salary: 480.87 }
]
==================
Person --> id=1, name=Isabel, salary=300.25
Person --> id=2, name=Martin Onate, salary=4000.35
Person --> id=3, name=Erick Revelo, salary=480.87
==================
P --> id=1, name=Isabel, salary=300.25
P --> id=2, name=Martin Onate, salary=4000.35
P --> id=3, name=Erick Revelo, salary=480.87
size of persons ArrayList -->13
Person --> id=1, name=Isabel, salary=300.25
Person --> id=2, name=Martin Onate, salary=4000.35
Person --> id=3, name=Erick Revelo, salary=480.87
Person --> id=1, name=Edison1, salary=1000
Person --> id=2, name=Edison2, salary=2000
Person --> id=3, name=Edison3, salary=3000
Person --> id=4, name=Edison4, salary=4000
Person --> id=5, name=Edison5, salary=5000
Person --> id=6, name=Edison6, salary=6000
Person --> id=7, name=Edison7, salary=7000
Person --> id=8, name=Edison8, salary=8000
Person --> id=9, name=Edison9, salary=9000
Person --> id=10, name=Edison10, salary=10000
dy456 ~\Desktop\ESPE2609-OOPSW36863\ws\cumbajin\U1\ws15CollectionsE
```